CARYA LABS

CARYA Network

CARYA Network Technical Whitepaper

TECHNICAL WHITEPAPER

Rooted in Value. Built to Grow.

ENGLISH

Technical specification for the CAA economic model, asset reserves, core mechanisms and ecosystem product matrix

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CAA / vCAA · ECOSYSTEM APPLICATIONS · CARYA CHAIN

CARYA NETWORK

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Project Description

We are CARYA Labs and the initiating team of CARYA Network (Walnut Network). We use this white paper to explain the project's product positioning, technical architecture, on-chain mechanism and ecological planning; the specific functions, parameters, contract addresses and online arrangements are subject to our officially released version and public on-chain records.

“We” in this article refers to the CARYA Labs and CARYA Network initiating teams. In addition to objective technical definitions, formulas and on-chain status, statements involving design choices, implementation plans, security commitments, governance principles and disclosure obligations all represent our position and commitment as the project sponsor.

We position CARYA Labs as an open value network laboratory and CAA as the ecosystem value layer of Walnut Network. CARYA Agent, a planned prediction market, a future native public chain and a Meme launch platform form the ecosystem product matrix. Protocol rewards, product-service revenue, developer settlement, balance-pool assets and ecosystem reserves are accounted for separately and disclosed through public rules and verifiable data.

Abstract

CARYA Network (Walnut Network) is an open value network initiated by CARYA Labs and planned for deployment on BNB Smart Chain. CAA serves as the ecosystem value and collaboration medium; vCAA, CTC, the balance pool, reward pool and progressive governance form the on-chain operating layer. CARYA Agent, the planned prediction market, the future CARYA Chain and the Meme launch platform are separate ecosystem directions connected to CAA utility. Our goal is not to rely on high emissions or initial liquidity to create short-term trading enthusiasm, but to allow real product use, user participation, developer contributions and on-chain economics to form a verifiable value cycle within one network.

CARYA Agent is a user-controllable AI agent system we built for individuals, teams and developers. We refer to industry practices for self-hosted personal agents and connect models, sessions, memories, tools, skills, and external services through a unified portal, enabling agents to understand tasks, invoke tools, and deliver results with explicit permissions. We do not claim that Walnut Network has affiliation, cooperation or compatibility with OpenClaw; unless we formally disclose it later, OpenClaw is only used as a public reference for self-hosted, multi-entry, tool-based agent architecture.

We set a fixed maximum supply of 310 million for CAA, do not set up private placement tradable chips, and define the initial market circulation as 0. We will allocate 290 million CAA in the ecological reward pool, 10 million in the balance pool, and 10 million as technology and operation construction shares and locked for 60 months of linear release. The assets in the destination reserve address are not automatically included in the tradable circulation because they have been minted. Only after real transactions, reward withdrawals or lock-up releases occur, the relevant assets will enter the corresponding circulation classification according to the definition of this white paper.

The CAA/USDT AMM liquidity pool is not established during the startup phase. The user buys CAA with U through the CTC/OTC order book; the opening reference price is 1.50 U = 1 CAA. The balance pool is a BAIQ-constrained quotation module that decides whether to open the sell side and its working belt based on the actual order, transaction, inventory, U-side reserve and destruction status. After the migration conditions are met, the protocol enters the V2 Manual Injection CAA/USDT LP and Built-in Swap phases.

Users can exchange CAA for 1: 1 within DApp to non-transferable vCAA. vCAA is used to record the effective principal and serve as a unit of account for static and dynamic rewards; after redemption, the static rewards are calculated according to the effective principal level, and the dynamic rewards are settled according to the performance of the effective community. Both types of rewards are billed as vCAA every 24 hours. The reward vCAA can be redeemed for CAA in the specified contract, or the automatic reinvestment can be transferred to the effective principal; the unreinvested reward will not be counted as the effective principal in the next cycle.

vCAA Principal or Bonus vCAA A staged transfer-out fee is charged when a liquid CAA is redeemed through a specified contract. The initial handling fee is 30%, which is reduced by 5 percentage points every 60 day (s) based on the agreement go-live time, with a minimum of 10%. The transfer-out fee for each stage is earned by the balanced pool and the operating address 50% respectively.

We summarize the long-term value logic of the Walnut Network into three parts: first, the accumulation of CAA quantity formed by user holdings and compound interest; second, the real ecological demand that may be formed by CARYA Agent usage, skill invocation, developer services and task collaboration; third, the possible changes in market prices under the condition that the real net demand continues to cover reward transfers, lock-up releases and market selling pressure. Time can participate in the cumulative calculation of the number of rewards, and product adoption can expand the usage scenarios of CAA, but neither can guarantee a price increase. Our responsibility is to provide real products, open rules, control authority, disclosure reserves and verifiable data, not to promise market results.

Chapter 1: Vision and Design Principles

1.1 Brand Origin

CARYA Labs takes its name from the pecan genus, and we thus establish "Walnut Network" as a user-oriented project brand. The walnut image contains a three-layer structure: the shell defines security and permission boundaries, the kernel preserves data and intrinsic value, and the seed represents the ability of agents, developers, and applications to continue to grow. We transform this structure into the design language of AI and open value networks:

- Each wallet is an independent participating unit;
- Smart contracts are public and verifiable rule boundaries;
- CARYA Agent is an intelligent interface between users and models, tools and ecological services;
- CAA and vCAA record the individual's value and participation status in the network;
- Recommended relationships form community connections, but cannot replace personal effective principal;
- The balancing pool provides liquidity base and ecological reserve;
- Skills, plug-ins and developer services form a scalable intelligent agent ecosystem;
- The 24-hour billing and 60-day charge cycles form an observable timescale.

Our brand expression is:

There are rules on the outside, but there is value on the inside; individuals are independent and grow together.

1.3 Eight Design Principles

Principle 1: Demand comes before circulation

The initial tradable circulation volume is 0. The balancing pool will only gradually release CAA under real buy orders and constrained quotations, and untransacted assets will continue to remain in the destination reserve address.

Principle 2: Personal capital comes before community rewards

Dynamic gear is determined by the performance of the active community. The dynamic reward is calculated by multiplying the effective community performance by the daily rate of the gear and distributed as vCAA. The community performance determines both the gear and the calculation base of the dynamic reward.

Principle 3: The number of rewards and the value of assets are disclosed separately

We calculate protocol rewards based on CAA and do not equate CAA rewards with USDT benefits. We also disclose the number of tokens, applicable fees, transaction prices and liquidity conditions in any earnings presentation.

Principle 4: Free entry and orderly exit

Users can deposit, withdraw or compound interest independently. The exit cost is disclosed in the contract and decreases according to certain rules as the protocol runs.

Principle 5: Limited reserves constrain unlimited commitments

Ecological rewards come from a limited reward pool. When the balance is insufficient, it will be weighted and distributed in proportion. When the balance is 0, there will be no settlement and no agreement arrears.

Principle 6: Market results are not guaranteed by narrative

We believe price depends on net demand, net supply and liquidity. We can improve supply and demand conditions and transparency through protocol design, but price increases will not be guaranteed.

Principle 7: Agents must be controlled by the user

AI agents can only run within the identity, data, tools and asset permissions granted by the user. Sensitive actions involving keys, signatures, transfers, transactions, deletions, releases or external sendings require clear authorization, quota limits and traceability logs by default.

Principle 8: Product use precedes token narrative

We base the ecological utility of CAA on real products and developer services. CAA can undertake access, collaboration, settlement, incentive and governance functions, but we will not replace product value with token incentives, nor will we force all AI services to use a single settlement asset.

Chapter 2: System Overview

2.1 Core Participants

Subject	Responsibilities
CARYA Agent User	Create and run individual or team agents, configure models, memories, skills and permissions
Agent Developer	Develop skills, plug-ins, workflows and professional agent services
Model and tool provider	Provides licensed access to model inference, API, data and execution tools
CAA Holder	Circulable for holding, transferring or trading CAA
vCAA Participants	Deposit CAA, receive vCAA, participate in rewards, withdraw or compound interest
Community Participant	Establish a non-recyclable recommendation relationship and form community performance
CTC Trader	Submit buy orders, sell orders, cancel orders and complete settlement during the startup phase
Ecological reward pool	Limited provision of static and dynamic rewards CAA Source
Balance Pool	Supports initial release, market quotation, transaction acceptance, reserve accumulation and later stage LP
Operating address	Receive a public proportion of the agreement fee for technology and ecological construction
Locked contract	Manage 60-month release of 10 million technical and operational shares
Protocol Level Market Agent	Generate constrained market analysis, quotation plans and execution boundaries, do not directly have unlimited asset permissions
Multiple signatures and time locks	Manage restricted parameters, exception pauses, and late AMM migrations
DAO	Take over some parameters and reserve governance rights in the mature stage

2.2 Operation phase

Phase A: Initiating trading with the order book

- The initial tradable circulation is 0;
- Does not create AMM LP;
- The user submits the purchase order through CTC;
- The balancing pool is gradually released according to real needs and constrained market plans CAA;
- CAA can enter the vCAA pledge system;
- Secondary buying and selling orders are formed between users;
- CARYA Agent Completed core runtime, permission system and first user testing.

Phase B: Market formation and reserve accumulation

- CTC Increased transactions between users;

- The balance pool accumulates USDT, CAA and reserve assets allowed by the protocol;
- Ecological rewards are gradually released based on the actual effective principal and community level;
- The protocol continues to disclose circulation, staking, rewards, reserves and concentration data;
- Open personal intelligence, the first batch of skills and developer access capabilities.

Phase C: AMM Migration

- The balancing pool USDT reaches the governance setting threshold;
- The market size, security and governance conditions meet the requirements;
- Create CAA/USDT LP;
- CTC Stop adding new orders and complete cancellation or settlement of existing orders;
- The user transfers to the protocol specified AMM/built-in Swap to complete the transaction;
- Open CAA within governance boundaries for agent service settlement and developer incentives.

Phase D: Reserves and Governance Expansion

- Gradually increase the transparency of reserve assets;
- When introducing mainstream assets such as BTC, ETH, disclose the native chain or mapped asset form respectively;
- Gradually transfer governable parameters to DAO;
- Establish mechanisms for proposals, voting, implementation delays and emergency checks and balances;
- Expand the intelligent agent technology market, multi-agent collaboration and third-party ecological applications.

2.3 Value and asset flow diagram

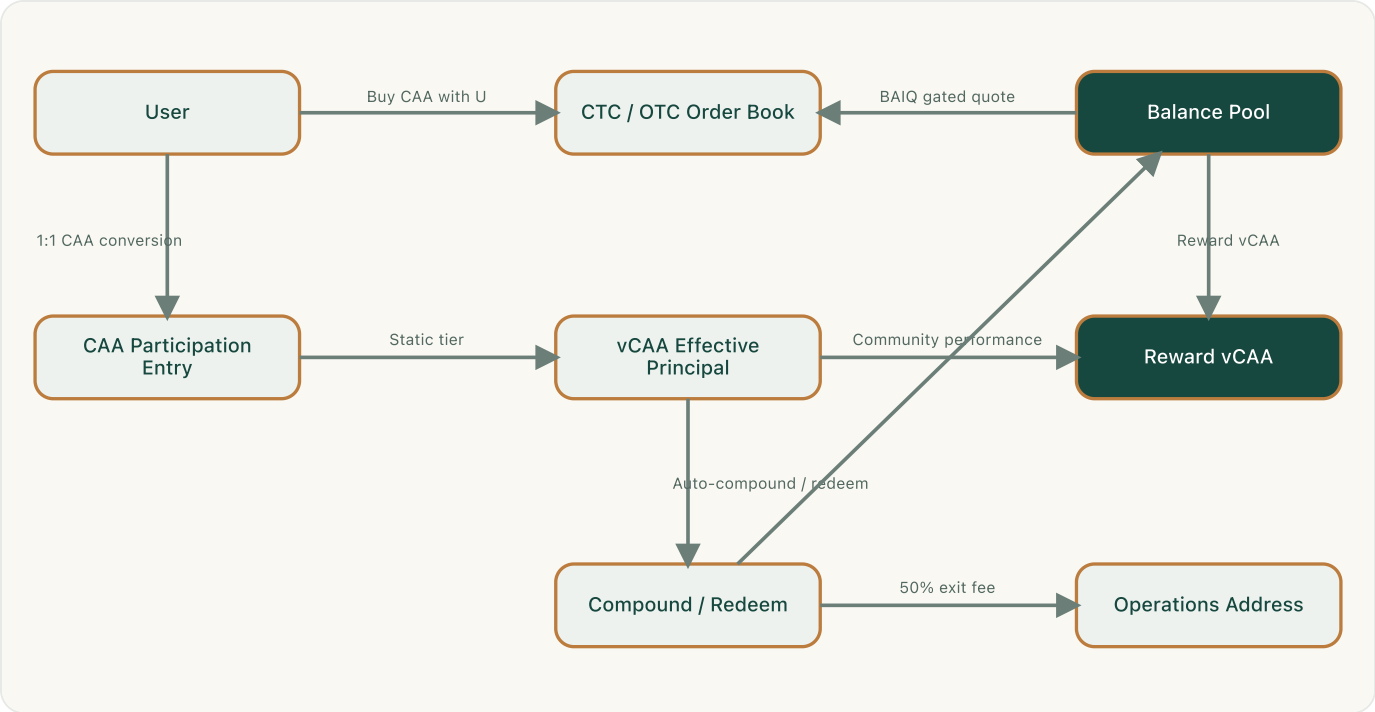


Figure 1 CAA / vCAA Value, Reward, and Exit Flow

Chapter 3: CARYA Agent Intelligent Ecology

3.1 Product Positioning

We are building CARYA Agent as a user-controlled AI agent system for individuals, teams and developers, and position it as one AI-agent direction within the broader ecosystem product matrix. It is not a single chatbot: it is an intelligent work gateway that connects models, memory, skills, tools and external services to complete multi-step tasks within user-defined permission boundaries.

Our product planning refers to industry architecture practices such as self-hosted personal agents, multi-model routing, skill plug-ins and multi-entry collaboration. We will give priority to supporting users' control over data, models, memories, tools and authorization scope; specific entrances, model suppliers and functions shall be subject to our officially released version.

3.2 Core Competencies

We plan the following capabilities for CARYA Agent:

1. Unified interactive portal: gradually support Web, desktop, mobile and audited message channels;
2. Persistent session and memory: Save task context in the workspace set by the user, and allow viewing, editing, exporting and deletion;
3. Multi-model routing: select different models according to task type, cost, delay, privacy and availability, and support fault degradation;
4. Skills and tool invocation: retrieval, analysis, content processing, workflow and authorized execution through versioned skills, plug-ins, API and on-chain connectors;
5. Task automation: supports scheduled tasks, event triggering, approval nodes, failed retries and result notifications;
6. Multi-agent collaboration: split complex tasks to agents with different roles and permissions, and summarize the results;
7. Developer extension: Provides SDK, skills list, test environment, measurement interface and release process;
8. Auditable operation: records models, tools, permissions, key input and output summaries and execution results to facilitate user review and security tracking.

We will not open all capabilities at the same time in the first version. For capabilities involving funds, external releases, sensitive data, or irreversible operations, we will enable them in phases after completing independent security testing.

3.3 Technical Architecture

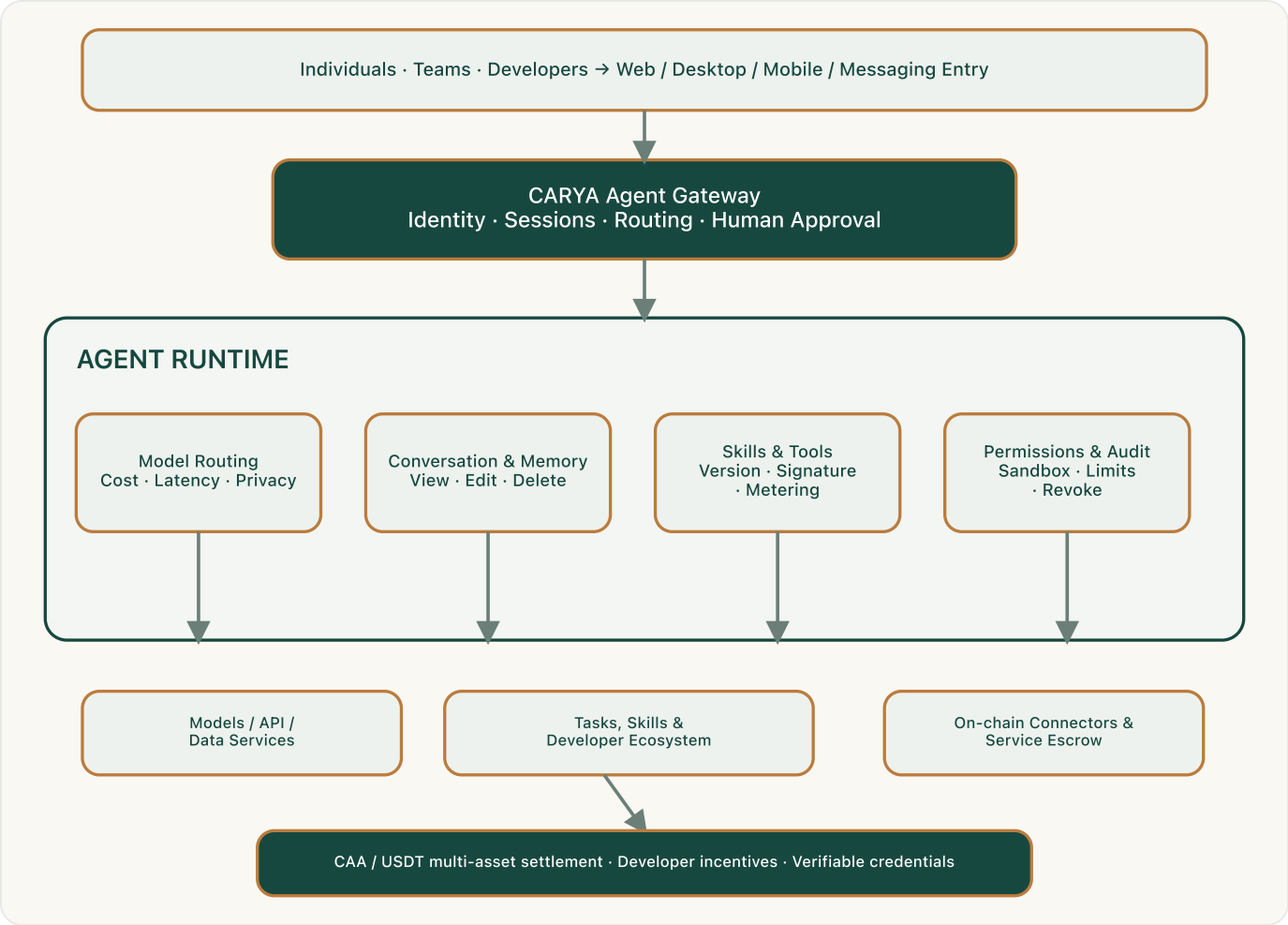


Figure 2 CARYA Agent System Architecture

We adopt a layered design of "off-chain intelligent execution + on-chain value and credentials":

- The interaction layer is responsible for user entrance, identity session, notification and manual approval;
- The agent operation layer is responsible for task planning, model calling, memory retrieval, tool orchestration and multi-agent collaboration;
- The security control layer is responsible for permission scope, sandbox, key isolation, quota, whitelist, audit and revocation;
- The ecological service layer carries skills, plug-ins, models, data, workflows and developer services;
- On-chain protocol layer records CAA/vCAA status, service settlement, custody, incentives, governance and necessary verifiable credentials.

In principle, large amounts of data, model reasoning and complete dialogue content are not directly written to the public chain. Only the address, hash, amount, time, status and authorization certificate required to complete settlement and audit are recorded on the chain, reducing privacy exposure and gas costs.

3.4 User Control and Permission Model

We adopt the principle of least privilege for CARYA Agent and set independent permission scopes for each agent, workspace, skill and tool, allowing users to view and revoke at any time. Our core controls include:

- Wallet private keys must not be uploaded to models, plug-ins or centralized agent services;
- On-chain operations preferentially use user wallet signatures, or use limited session authorization with deadlines, quotas, assets and contract whitelists;
- Transfer, transaction, authorization, deletion, public release and sending data to external accounts, manual confirmation is set by default;
- Sensitive skills configuration single and periodic quota, rate limit, destination address whitelist and emergency stop;
- The model only obtains the minimum context required to complete the current task, and sensitive information is stored in isolation, desensitized or processed locally;
- Each critical execution forms a user-readable log and allows subsequent permissions to be revoked, rather than interpreting a single authorization as a permanent authorization.

We will not let CARYA Agent become the custodian of user assets by default, nor will we use AI judgment to replace user signatures. If account abstraction or restricted automatic execution is introduced in the future, we will make the permission format, recovery process, guarantee boundaries and independent audit results public.

3.5 The role of CAA in the intelligent agent ecology

We position CAA as an open collaboration medium for the CARYA Agent ecosystem, rather than the only threshold for using all functions. After we complete product release and governance confirmation, CAA can assume:

1. Product access: Redeem or offset specific advanced capabilities, computing resources, storage, professional skills and subscription benefits;
2. Task settlement: Complete verifiable task settlement between users and developers, skill providers, data or tool service providers;
3. Developer incentives: Support developer growth based on real calls, user retention, quality evaluation and security records;
4. Service Guarantee: Specific services can lock CAA as a quality assurance or dispute handling deposit;
5. Collaboration and governance: Participate in ecological proposals, parameter voting, resource budgeting and service rule governance;
6. Identity and quota: After the governance is clarified, the CAA or vCAA status can be used for partial quotas, test qualifications and ecological levels, but does not automatically generate undisclosed rights.

To reduce the impact of token fluctuations on product availability, we allow some models, API, Gas, and third-party service costs to be denominated in USDT, BNB, fiat currency, or other governance-approved assets. We will clearly display whether CAA must be used, its scope of application, price conversion and fee cap on the corresponding product page.

3.6 Skills, tasks and developer market

We plan to establish an open but governance-bound skills and task market for the Walnut Network:

- Developer releases skill package with version number, permission statement, dependencies, price and signature;
- The platform displays the skill calling scope, data destination, cost, security label and historical version;
- After user authorization, the agent calls skills according to tasks and generates verifiable measurement and result receipts;
- For tasks that can be objectively accepted, the on-chain custody contract can be used to complete submission, confirmation, refund and dispute handling;
- Service revenue can be allocated between developers, model/API costs, infrastructure and protocol services, with specific proportions subject to product rules and governance proposals;
- Malicious skills, false calls, plagiarism, vulnerabilities and data abuse will trigger delisting, suspension, deposit disposal or other public rules.

We prioritize ecological incentives based on real usage quality, rather than purely on registration volume or token holdings. The core metrics we use include number of successful tasks, reuse rate, user retention, failure rate, chargeback rate, developer revenue, and security incident records.

3.7 Integration with on-chain mechanisms

We form the following closed loop through the CARYA Agent and CAA protocol layers:

1. Users complete real tasks through intelligent agents, generating demand for models, skills, tools and developer services;
2. Users can choose CAA or other supported assets for payment, and the usage of CAA forms a discloseable ecological demand indicator;
3. Developers receive income or governance-approved incentives for quality services to continue building skills and applications;
4. Part of the disclosed protocol service revenue can replenish the infrastructure, balancing pool, developer plan or security fund according to the governance rules;
5. vCAA Participation, community collaboration and governance provide a long-term user base for the ecosystem, but protocol rewards and product revenue must be accounted for and disclosed separately;

6. Product usage data is only aggregated within the scope permitted by user authorization and privacy rules, and token incentives are not used to induce unnecessary data authorization.

The key to this closed loop is not to use financial mechanisms to subsidize ineffective calls, but to allow product use to generate service value, and let the service value reversely support developers, infrastructure and network collaboration. Product demand may improve the usage base of CAA but does not constitute a guarantee of a price increase.

Chapter 4: Token Issuance and Supply Classification

4.1 CAA Basic parameters

Parameters	Value or status
Network	BNB Smart Chain
Standard	BEP-20 / ERC-20 compatible interface
Name	CAA
Accuracy	18
Maximum Supply	310,000,000 CAA
Private placement tradable distribution	0
Initial market circulation	0
Subsequent issuance	does not allow exceeding the maximum supply; we plan to permanently close minting permissions after the one-time minting of Genesis

BNB Smart Chain supports EVM compatible smart contracts, so we adopt ERC-20 compatible interface for CAA and use the mature EVM tool chain for development and verification.

4.2 Initial Allocation

Allocation Purpose	Quantity	Proportion	Initial circulation classification
Ecological reward pool	290,000,000	93.5484%	Non-tradable
Balance Pool	10,000,000	3.2258%	The untransacted portion is not included in the tradable circulation
Technical and operational lockup	10,000,000	3.2258%	The lock-up period is not included in the tradable circulation
total	310,000,000	100%	—

"Initial 0 circulation" does not mean that the token has not been minted yet, but it means that there are no CAA that have entered the public wallet and can be freely traded. The creation allocation is all located in the purpose contract or locking address, and changes status according to public rules.

4.3 Technology and Operation Lockup

We have locked 10,000,000 CAA technical and operational shares through PinkLock for a total of 60 periods, with a release every 30 days. PinkLock position locking address, beneficiary address, start time, release amount and actual collection record are disclosed before the main network is launched. The baseline releases per period are:

V_month=floor((10,000,000×10^18)/(60))

The smallest unit remainder produced by integer division is released in the last period. We will disclose the locking address, benefit address, starting time, release amount in each period and actual collection record. Changing the beneficiary address must go through multi-signature and time lock, and the total amount, starting time and remaining release curve must not be changed.

4.4 Supply terms

In order to avoid the confusion between "minted amount" and "market circulation amount", the agreement uses the following definition:

- Maximum supply (Max Supply): the theoretical upper limit allowed during the life cycle of the contract;
- On-chain Total Supply: ERC-20 totalSupply() return value;
- Black hole balance (Dead Balance): transferred to CAA, a publicly unavailable address;
- Effective Supply: the total supply on the chain minus the black hole balance;
- Tradable Circulation: CAA that can be freely transferred or traded from a public address;
- Staked Supply: CAA that has entered the pledge account and is represented by vCAA;
- Reserved Supply: Reward pool, untransacted balance pool balance and unlocked construction share.

If the destruction is realized through black hole transfer, then:

S_effective=S_total-B_dead

The tradable circulation volume is estimated according to the following formula:

C_t=C_t-1+R^bp_t+W^reward_t+U^vest_t-D^stake_t-B^market_t

where:

- R^bp_t: The actual transaction volume released by the balancing pool to the market;
- W^reward_t: The net tradable amount formed by reward withdrawal;
- U^vest_t: The number of locked positions released in the current period and entered into tradable addresses;
- D^stake_t: The amount deposited into the pledge account from the tradable state;
- B^market_t: The market circulation that enters the black hole after the balancing pool is taken over CAA.

4.5 Ecological utility boundary of CAA

Our goal in designing CAA is to connect the chain of Walnut Network to participate in product collaboration with CARYA Agent. Its potential functions include product access, agent task settlement, developer incentives, service deposits and governance, but we will only enable relevant functions after the corresponding functions are officially released, and clearly disclose the pricing methods, fees and user rights.

CAA does not represent CARYA Labs equity, profit sharing rights, or ownership of balancing pool assets. Holding CAA or vCAA does not automatically obtain AI services, income, voting rights or priorities; unreleased product plans shall not be included in the user's current rights and interests. The product side retains multi-asset payment capabilities to avoid CAA price fluctuations blocking normal services.

Chapter 5: CAA and vCAA account models

5.1 Properties of vCAA

vCAA is a non-transferable nominated contractual journal voucher, is not a separate market asset, and is not entered into CTC or AMM. It distinguishes between the user's valid principal and the reward vCAA: the valid principal is used for the static position, and the reward vCAA is used to record the static, dynamic reward that is settled every 24 hours.

vCAA does not open ordinary transfer, approve and transaction functions; only designated contracts are allowed to complete 1: 1 exchange, reward settlement, automatic reinvestment and redemption, regardless of whether a non-transferable interface or internal ledger is used. If tokenization vCAA is implemented for the wallet presentation, all normal transfer functions will be disabled by default.

5.2 Deposit

The user must redeem a minimum of 100 CAA, which must be an integer multiple of 1 CAA, and can be called:

```
stake(amount, referrer)
```

Execution result:

1. User CAA transferred to the pledge vault;
2. The user’s effective principal is increased by the same amount vCAA;
3. Bind the referral relationship and pay the 0.0005 BNB binding fee when participating for the first time;
4. Principal changes take effect from the next full 24-hour period;
5. triggers necessary community performance snapshot updates.

We will not automatically terminate the recommendation relationship due to subsequent withdrawals by users. Regarding whether to allow migration of superiors, our default design is not to allow cross-region double calculations and rewriting of historical performance.

5.3 Reward status

Each user maintains at least the following status:

```
principalVcaa
pendingStaticVcaa
pendingDynamicVcaa
lastSettledEpoch
staticTier
dynamicTier
referrer
directChildrenCount
```

The static reward vCAA to be claimed is recorded separately from the dynamic reward vCAA to be claimed. The two do not automatically increase the principal; the user can choose to redeem it in the specified contract, or choose to automatically reinvest it into the effective principal.

5.4 Withdrawal and Compound Interest

Users can handle static or dynamic rewards respectively:

```
claimStatic(amount)
claimDynamic(amount)
compoundStatic(amount)
compoundDynamic(amount)
```

Extract

Withdrawal means that the reward vCAA will be redeemed back to the liquid CAA in the specified contract, and the transfer-out fee will be deducted according to the current global stage. The user receives the net amount, and the handling fee is 50% from the balanced pool and the operating address.

Compound Interest

Automatic reinvestment does not occur from vCAA to CAA, so there is no transfer fee. The reinvestment amount is transferred from the reward vCAA to the effective principal and participates in the static range and effective cell performance calculation from the next settlement cycle.

Principal withdrawal

Users can actively reduce or redeem the vCAA principal in full. The same staging fee applies when switching back to CAA. The remaining principal will be re-matched to the static position from the next cycle; after all withdrawals, no new static rewards will be generated, and the effective community performance will also be updated according to the rules.

Chapter 6: 24-hour reward engine

6.1 Time Model

Suppose the protocol launch time is T_0, and a reward period is 86,400 seconds. The cycle number is:

epoch(t)=floor((t-T_0)/(86400))

Changes in principal, compound interest and community performance will take effect from the next full cycle, preventing users from instantly increasing their principal before the settlement boundary and obtaining full cycle rewards.

We adopt lazy settlement: instead of performing an on-chain cycle for all addresses every day, the system calculates rewards based on snapshots and unsettled periods during user interaction, Keeper calls, or batch settlement. This can avoid unbearable gas consumption when the number of participating addresses increases.

6.2 Static reward level

Valid vCAA The principal reaches	Daily Static Reward Rate in vCAA
100	0.10%
500	0.15%
1,000	0.20%
3,000	0.23%
5,000	0.25%
10,000	0.28%
30,000	0.30%

The static gear function is:

r_s(P)=max{r_j | P≥ q_j}

The theoretical static reward for a single period is:

E^s_u,e=P_u,e× r_s(P_u,e)

is less than 100 vCAA, the static reward rate is 0.

6.3 Effective community performance

Each direct subordinate address constitutes an independent zone, and the effective principal of the address and its complete subordinate network constitutes the performance of the zone. Each address can be bound to a maximum of 256 direct reports.

Suppose user u has n direct areas, and the area performance is:

$Z_{u,1}, Z_{u,2}, \ldots, Z_{u,n}$

The largest area is:

$Z^{max}_u = \max_i Z_{u,i}$

The effective community performance is:

$S_u = \Sigma_{i=1}^{n} Z_{u,i} - Z^{max}_u$

We only subtract one maximum area from the protocol, add all the remaining areas, and do not perform additional performance deduplication. The recommendation graph must satisfy a single parent node and an acyclic structure, so different direct areas naturally belong to different subtrees.

6.4 Dynamic reward level

Active Residential Performance Reached (CAA)	Daily Dynamic Incentive Rate (based on active community performance)
10,000	0.10%
50,000	0.20%
100,000	0.23%
300,000	0.25%
500,000	0.28%
1,000,000	0.30%
3,000,000	0.33%
5,000,000	0.35%
10,000,000	0.38%

Start the status reward after the effective community performance reaches the first tier of 10,000 CAA. The dynamic gear function is:

$r_d(S) = \max\{d_j \mid S \geq s_j\}$

Dynamic rewards are directly calculated based on the performance of the active community:

$E^{d}_{u,e} = S_{u,e} \times r_d(S_{u,e})$

6.5 Static and dynamic overlay

The two types of rewards are calculated separately:

$E_{u,e} = E^{s}_{u,e} + E^{d}_{u,e}$

Static and dynamic rewards are awarded in vCAA. Static rewards are based on the effective principal of the individual, and dynamic rewards are based on the performance of the effective community, so the two cannot be simply added to the fixed daily rate of return of the individual.

$$E_{u,e} = P_{u,e} \times r_s(P_{u,e}) + S_{u,e} \times r_d(S_{u,e})$$

The award tier and formula describes how vCAA is calculated and does not constitute a fiat yield or price commitment. The actual settlement is affected by the balance of the reward pool, whether it is automatically reinvested, transfer-out fees, market transactions and liquidity.

6.6 Scalable settlement architecture

Dynamic Volume involves a multi-tiered recommendation subtree. If each redemption, redemption, or re-investment is updated along an infinite ancestral path, there will be uncontrollable gas costs. Therefore, a hybrid architecture of "on-chain asset custody + on-chain event data + periodic verifiable settlement" is adopted:

1. All deposits, withdrawals, compound interest and changes in recommendation relationships generate on-chain events;
2. The independent indexer calculates a valid principal snapshot, valid cell performance, and theoretical reward vCAA for each cycle based on public events;
3. Settlement coordinator submission cycle Merkle Root, total theoretical reward amount and weighting coefficient;
4. The user redeems the reward vCAA or performs an automatic replay through Merkle Proof;
5. The total allocation amount shall not exceed the available balance of the reward pool for the current period;
6. Root, input block range, algorithm version and output file hash are public;
7. The mature stage allows multiple settlers to submit results and introduces deposits, challenge periods and error penalties.

This architecture allows any third party to recalculate the results based on on-chain data while avoiding full address cycles blocking settlement.

Chapter 7: Reward Pool Balance Constraints

7.1 Theoretical rewards and actual rewards

Suppose the total theoretical reward of all users in period e is:

$E_e=\Sigma_{u=1}^{N} E_{u,e}$

The available balance of the ecological reward pool at the time of cycle settlement is B_e. Define weighting coefficient:

$$\alpha_e=\begin{cases}1, & B_e \geq E_e \\ \frac{B_e}{E_e}, & 0<B_e<E_e \\ 0, & B_e=0\end{cases}$$

The actual reward for users is:

$A_{u,e}=E_{u,e}\times\alpha_e$

7.2 Insufficient balance and zero balance

- When the balance is sufficient, the user will receive the full theoretical reward;
- When the balance is insufficient, all users will be distributed in the same proportion according to the theoretical reward weight;
- When the balance is 0, no rewards will be settled in the current period;
- The unsettled portion does not form future claims or agreement arrears;
- The smallest unit remainder generated by rounding is retained in the reward pool and enters the next cycle.

7.3 Emission Monitoring

We will display in the public dashboard at least:

- Current Static Theory Award vCAA;
- Current Dynamic Theory Award vCAA;
- Actual weighting coefficient;
- Actual allocated amount;
- Remaining balance of the reward pool;
- Remaining period estimated based on average emissions over the last 30 days;
- Stress test under different user growth and compound interest assumptions.

Chapter 8: Transfer Fees and Exit Paths

8.1 Rate function

The initial transfer fee is 30%, which will decrease by 5 percentage points every 60 days starting from the global launch time of the protocol, with a minimum of 10%.

Suppose the current time is t, and the number of completed cycles is:

$k(t)=floor((t-T_0)/(60\times86400))$

The handling fee is:

$f(t)=max(30\%-5\%\times k(t),10\%)$

Protocol running time	Transfer fee	Balance Pool	Operating address
Day 0-59	30%	15%	15%
Day 60-119	25%	12.5%	12.5%
Day 120-179	20%	10%	10%
Day 180-239	15%	7.5%	7.5%
After day 240	10%	5%	5%

8.2 Transfer out calculation

When a user redeems X vCAA principal or reward vCAA:

$CAA_user=X\times(1-f)$

$CAA_balance=X\times(f)/(2)$

$CAA_operation=X\times(f)/(2)$

The transfer fee is not destroyed. Related CAA enters the public balancing pool address and operating address respectively.

8.3 Complete realization cost

User experience from vCAA monetization to U at least:

1. vCAA principal or reward vCAA converted to CAA in the specified contract;
2. CAA Sold on CTC or later on AMM.

When the 30% transfer-out fee corresponds to 100 vCAA redemption, the user first gets 70 CAA; whether the seller's transaction fee is generated when the CTC/OTC is sold subsequently, and the calculation is based on the CTC rule.

$100 \times (1 - 30\%) = 70$

This example only illustrates the net amount of CAA in the redemption phase; the transaction price, closing and subsequent CTC fees are not within the illustrated caliber of this 70 CAA.

Chapter 9: CTC Order Book

9.1 The role of CTC

CTC/OTC is an on-chain order book trading system for the start-up phase of the walnut network, which is used to buy CAA with U, form real orders and make real transactions; unfulfilled orders can be reversed. The startup period has no initial AMM LP.

CTC/OTC opening reference price is 1.50 U = 1 CAA. Actual closing still comes from user orders, balanced pool constrained quotes and true matching; BAIQ's reference price is only revised when the gate passes, and does not constitute a fixed price, insured price or repurchase commitment.

9.2 Basic Rules

Parameters	Rules
Priced Assets	Agreement specifies BSc USDT (Opening reference price 1.50 U/1 CAA)
Each order fee	0.0001 BNB
CAA Minimum sell order quantity	100 CAA
CAA Sell order increment unit	1 CAA integer multiple
U Minimum number of buy orders	Convert at least 100 CAA at the price of the order
U Purchase order increment unit	Integer multiplier corresponding to 1 CAA
Buyer’s transaction fee	5%
Seller’s closing fee	5%
Handling fee distribution	2.5% of each side fee enters the balance pool and 2.5% enters the community incentive
Unfilled orders	can be revoked
BNB Order fee	Balance Pool Whitelist Market Making Orders Waived Hanging Order Fee and Transaction Fee; Other Orders Paid and Not Refunded

9.3 Settlement assets and handling fees

CTC Bilateral fee settlement and retention of no pending order fee, no turnover fee rules for balanced pool whitelisting orders:

- The buyer’s fee is deducted from its receivable CAA;
- The seller’s fees are deducted from its receivable USDT;
- In the buyer's 5% CAA fee, 2.5% enters the balance pool and 2.5% enters the community incentive;
- In the seller's 5% U fee, 2.5% enters the equilibrium pool and 2.5% enters the community incentive.

If the order is filled with Q pieces of CAA and the price is p USDT/CAA, then the nominal transaction amount is:

$$N = Q \times p$$

The buyer actually received:

$$Q_{buyer} = Q \times 95\%$$

The seller actually received:

$$N_{seller} = N \times 95\%$$

9.4 Anti-Dump Index

The Anti-Dump Index uses the first transaction at 08:00 each day as the day's reference price. When the market price declines from that reference by one of the thresholds below, the applicable transaction fee is settled at the corresponding total rate.

Price decline	Total fee	Designated address	Direct CAA burn
10%	20%	8%	12%
15%	30%	8%	22%
20%	50%	8%	42%

The fixed 8% goes to the designated address. Every fee increment triggered by the Anti-Dump Index above that fixed 8% is directly burned as CAA during settlement; it does not enter the designated address.

If one transaction meets more than one decline threshold, the highest triggered tier applies. This index describes fee and burn routing only; it is not a price-protection, price-guarantee, buyback, or liquidity commitment.

9.5 Matching Principles

We use the following matching principles for CTC:

1. Price priority;
2. Same price and time priority;
3. supports partial transactions;
4. Order assets are pre-escrowd;
5. The order uses a non-repeatable nonce;
6. Canceling an order only releases the untransacted balance;
7. Transactions, order cancellations and fees all trigger on-chain events.

In terms of technical implementation, we use discrete price ticks and FIFO queues to reduce on-chain sorting costs. The matching Keeper can only execute transactions within the order limit range and cannot cross the price boundary signed by the user.

9.6 CTC Economic Impact of Fees

In the transaction fee of the buyer and the seller 5%, each side is allocated according to the 2.5% balance pool/2.5% community incentive; the balance pool whitelist market making order is free of hanging order fee and transaction fee. For ease of checking, we continue to disclose:

- Nominal transaction price;
- Buyer’s cost included;
- The seller actually received the account;
- Effective bid-ask spread;
- Balancing pool and operating address accumulated fees.

Chapter 10: Protocol Level Market Intelligence

10.1 Positioning

BAIQ (Burn-Adjusted Inventory Quoter) is a constrained market capitalization quoting module called by the balance pool, not an unbiased market maker. It only schedules quotation conditions and does not schedule CAA quantities; it is used to form a one-way ratchet reference quotation based on inventory, U-side reserves, actual transactions, market circulation and destruction status.

BAIQ does not have unlimited pricing, unlimited pending orders, or unlimited access to Balanced Pool assets. It can only submit quotes within the four-door control, daily reference price revision, inventory and reserve boundaries of the contract; quotes are still subject to review by the contract, Keeper, multi-signature and time lock.

10.2 Input data

- CTC Buy order, sell order and order depth;
- Trading volume, average trading price and volatility;
- Net buying and net selling;
- Tradable circulation and vCAA pledged amount;
- Reward withdrawal and compound interest amount;
- Balance pool CAA, USDT and other reserves;
- Concentration of large addresses;
- Rewards to be released and lock-up plan;
- Abnormal orders, order cancellations and address association signals.

10.3 Output data

- Target buying and selling prices;
- Target bid-ask spread;
- Order quantity plan;
- Maximum fund usage in a single day of the balancing pool;
- Running status level;
- Whether to trigger manual review or circuit breaker.

We will display the desensitization strategy, backtesting results and running status of market agents in the CARYA Agent data center for users to query; we will not open unauthorized order identities, private strategies or vault keys to ordinary user agents.

10.4 Constrained Execution

We require that AI output is always constrained by on-chain parameters:

- Daily reference price increase δ: when the gate is passed, the normal day is 0.10% — 0.20%, the strong absorption day is 0.30% — 0.40%, and the extreme day is from 0.50%; when the gate is not passed, δ = 0;
- Maximum USDT usage ratio in a single day: maxDailyReserveUsage(TBD);
- Minimum USDT Reserve Ratio: minReserveRatio(TBD);
- Maximum single order quantity: maxOrderSize (TBD);
- Quotation validity period: quoteTTL (TBD);
- Abnormal fluctuation suspension threshold: circuitBreakerThreshold (TBD).

The BAIQ packet contains the input block range, net U, inventory skew, destruction intensity, inventory depth, δ, reference price, policy version, output, timestamp, and hash. Restricted Keeper or multi-signature accounts submit a quote, which is executed after the contract is verified with four gates and hard boundaries; no model, Keeper or operating account shall bypass reserve permissions, rate caps, destruction stop lines or governance time locks.

Chapter 11: Balance Pool and Ecological Reserve

11.1 Functional Boundaries

Balance pool responsibility:

- Startup period CAA Gradual release;
- CTC Market quotation and limited transaction acceptance;
- Record the CAA and U traffic of the transaction fee and redemption fee, and supplement the U-side quoting capacity, community incentives, and operational address of the balance pool according to the established routes;
- Later CAA/USDT LP construction;
- Establish a public ecological reserve after governance allows;
- Perform restricted liquidity operations under extreme conditions.

The balancing pool is not responsible for:

- Fixed price buyback commitment;
- Unlimited USDT Redeem;
- User principal guarantee;
- Ordinary asset withdrawal without governance;
- Unlimited domination of AI or single operating account.

11.2 Asset Permissions

We deploy the balancing pool as a treasury contract instead of a normal single-signature wallet. Asset expenditures must be for whitelisted purposes:

- CTC Transaction settlement;
- Restricted market quotation;
- AMM LP injection;
- Governance-approved reserve allocation;
- Audit, migration or security incident handling.

We jointly control non-routine operations through multi-signature and time lock. Individual withdrawals of assets are not allowed from emergency suspended accounts.

11.3 Reserve Disclosure

The data we disclose includes at least:

- Balance Pool CAA;

- BSC USDT;
- Other BSC assets;
- Cross-chain reserve address and asset form;
- Order assets have been locked;
- Available reserves;
- 30-day net inflow and net outflow;
- Market acceptance of usage;
- AMM LP positions and LP ownership.

If BTC, ETH, SOL and other assets are allocated, we will clearly disclose whether they are native chain assets, cross-chain mapping assets or centrally issued anchor assets. Different forms have different custody, bridging and price anchoring methods and we do not consolidate disclosures by name only.

11.4 BAIQ Market Cap Module

BAIQ (Burn-Adjusted Inventory Quoter) is a constrained market cap quoting module called by the balance pool: not an unbiased market maker, but a one-way ratchet with a supply contraction constraint. It does not schedule the CAA quantity, only the quotation conditions and whether the sell side is open.

The selling side is only opened when the four gates are met at the same time: the net U (F) is positive on the day, the inventory skew (S) still retains the U reserve, the destruction intensity (B) synchronous shrinkage supply, and the opening depth is sufficient to undertake this expansion. When any one of the conditions is not met, the selling side is closed, or only the very thin quotation is retained; $\delta = 0$ at this time.

The reference price of BAIQ is revised upward in the gated passing gear without returning to yesterday's reference price; it is not a price protection commitment. Conventional, strong absorption and extreme working bands correspond to δ 0.10% — 0.20%, 0.30% — 0.40% and 0.50%. Based on the opening 1.50 U, the reference price of the illustrated scenario of 0.10%/0.30%/0.50% is about 2.15 U/4.41 U/9.03 U if it passes through the gating every day for 360 consecutive days; the illustration is a continuous gating scenario, not a daily compound interest or an actual transaction commitment.

11.5 Asset Reserves and Ecosystem Coordination

The balance pool and ecosystem reserves are neither the reward pool nor a market-price guarantee. The 10 million CAA balance-pool allocation, USDT reserves created through trading and other governance-approved assets are accounted for separately by address and purpose. Unsold CAA remains a purpose-specific reserve and does not automatically enter market circulation.

Under disclosed permissions and governance conditions, reserves may support constrained quotation and settlement buffers, a later CAA/USDT LP migration, ecosystem-service infrastructure and approved developer or application support. Every use is subject to allowlists, multisig control, limits, timelocks and on-chain disclosure, and may not bypass reward-pool balances, burn stop lines or circulation classifications.

Reserve size, composition, available ratios, cross-chain addresses, LP ownership and related transactions will be disclosed continuously. Reserves provide execution capacity and risk buffers only; they do not constitute buyback, price-support, return or liquidity commitments.

Chapter 12: Destruction and Stop Line

12.1 Destruction Definition

We currently use the black hole address record to permanently exit the use of CAA. Blackhole transfers do not automatically change the standard ERC-20 totalSupply(), so we use "effective supply" to express supply reductions.

CAA that has been transferred to the black hole cannot be recovered and will not be reallocated after reaching the stop line.

12.2 Destroying sources at the current stage

We have determined the following startup path in this article, which will ultimately be subject to our public deployment parameter table and audit code:

1. After the balance pool directly accepts the CAA sold by the user, the received CAA is transferred to the black hole and will not be re-listed;
2. After the ordinary user completes the sales order to other users through CTC, the ecological reward pool synchronously transfers into the black hole according to the actual number of CAA transactions; the synchronous destruction ratio defaults to 100% and can be adjusted within 0% — 100%;
3. The synchronous destruction of the reward pool when the sale order is closed follows the burnSyncRatio: burnAmount = number of deals CAA × burnSyncRatio. This ratio defaults to 100%, and its manageable range is 0% — 100%; it is not adjusted downward in proportion to the size of the market.
4. The vCAA redemption fee is not a destruction, and still press 50%/50% to enter the balance pool and operating address.

12.3 Stop Line

- When the effective supply of CAA drops to 100,000,000, stop reducing the effective supply of the entire network;
- When the releasable balance of the ecological reward pool drops to 10,000,000 pieces, the synchronous destruction of the balance of the ecological reward pool will stop;
- After reaching the stop line, the corresponding destruction action will stop itself, no deduction will be made, no transfer will be made to other addresses, and no additional fees will be incurred;
- The two stop lines are judged independently.

12.4 Destruction does not equate to price increase

Destruction reduces current circulation or potential future supply, but price still depends on demand and liquidity. If net demand falls faster than supply, prices could still fall. Therefore destruction can only be disclosed as a supply variable and not as a price guarantee.

Chapter 13: AMM Migration

13.1 Triggering Principle

The startup phase does not establish AMM. After the conditions of CTC/OTC and BAIQ are met, the balance pool is manually injected into CAA and U according to the V2 process to establish LP and open the built-in Swap; the specific migration time is confirmed by the public process.

In addition to reserve size, we also require:

- CTC has formed a continuous transaction record;
- The tradable circulation volume reaches the minimum requirement;
- The number and concentration of holders meet governance standards;
- AMM Contract and routing completed third-party audit;
- The initial LP scale can cover the expected transaction slippage;
- The ownership and management rights of LP Token have been transferred to the black hole address;
- Stablecoins, oracles, MEV and emergency solutions have been tested.

13.2 Migration Process

1. Governance release AMM Migration proposal and parameters;
2. Enter announcement and execution delay period;
3. CTC Stop accepting new orders;
4. Uncompleted orders are allowed to be cancelled;
5. Follow the V2 process to inject CAA and U manually;
6. Ownership and management rights of the LP Token are transferred to the black hole address;
7. Open built-in Swap; BTC, ETH, BNB and other mainstream currencies obtained with built-in Swap are not transferable;
8. CTC only retains the historical query and inventory settlement functions.

13.3 AMM Handling fee

Late AMM/Built-in Swap fees, quotes, and asset boundaries are exposed with V2 parameters. LP Ownership and management rights have been transferred to the black hole address, and the mainstream coins obtained by the built-in Swap are not transferable; any adjustment shall not be done instantly by a single address, and the settings are:

- Contract hard limit;
- Multiple signatures or DAO approval;
- Execution time lock;
- Parameter change event on the chain;

- Announcement period before taking effect.

Chapter 14: Long-term Value and Supply and Demand Models

14.1 The number of rewards is not the final value

The user's realizable value at time T can be abstracted as:

$V_T=Q^{liquid}_T\times P_T+V^{realized}_T-C^{acquire}-C^{fee}-C^{gas}$

where:

- Q^{liquid}_T: The quantity of CAA that can be circulated and transacted;
- P_T: actual transaction price;
- $V^{realized}_T$: previously realized transaction value;
- $C^{acquire}$: Cost of acquiring CAA;
- C^{fee}: transfer-out, CTC or AMM fee;
- C^{gas}: Gas cost on the chain.

The reward increases by Q, and the market supply and demand determines P. Both may increase at the same time, or one may increase and the other may decrease.

14.2 Correct expression of daily reward rate

Static rewards are calculated according to the effective principal level, and dynamic rewards are calculated according to the effective community performance; both are issued as vCAA. The static daily rate range is 0.10% — 0.30% and the dynamic shift daily rate range is 0.10% — 0.38%, but due to different calculation bases, the two items cannot be expressed as individual fixed 0.68% daily rate of return.

Therefore, we will not promote Walnut Network as an "absolutely low-yield agreement." The exact positioning we use is:

does not simply pursue higher short-term emissions than the market, but puts incentives, circulation, reserves, exit and real demand into the same long-term constraint model.

14.3 Time accumulation

If the user chooses compound interest, the change in principal is a piecewise function:

$P_{e+1} = P_e + (A \hat{} s_e + A \hat{} d_e) \times c_e$

where $c_e \in [0, 1]$ indicates the proportion at which the user will select an automatic repeat vote for the current reward vCAA. After automatic reinvestment, the amount enters the effective principal from the next cycle; as the principal crosses the gear, the static reward rate may change in the next cycle.

time can accumulate CAA quantity, but at the same time increase reward pool emissions and potential future selling pressure. So we will observe both:

- Compound interest rate;
- Reward withdrawal rate;
- New pledge amount;
- Add real buy;
- Reward pool balance;
- Market liquidity.

14.4 Necessary conditions for price support

Define periodic net demand:

D^net_t=D^participation_t+D^agent_t+D^secondary_t-D^sell_t

Among them:

- D^participation_t: Net demand for on-chain participation;
- D^agent_t: Net demand due to CARYA Agent product access, task settlement, developer services or deposits;
- D^secondary_t: Net purchases in the secondary market that do not fall into the first two items;
- D^sell_t: current selling demand.

We base the three types of requirements on verifiable data classification to avoid double counting of the same purchase. For payments for agent services done in USDT or other assets, we will not fictionalize them as CAA requirements.

Define new tradable supply:

S^new_t=R^bp_t+W^reward_t+U^vest_t-D^stake_t-B^market_t

When other conditions remain approximately unchanged, the necessary conditions for price to obtain sustained support are:

D^net_t>S^new_t

This inequality is not a sufficient condition. Market depth, position concentration, external market conditions and trading expectations may still change price results.

14.5 Supply and Demand Scenario

Growth Scenario

- New real buy continues;
- CARYA Agent Continued growth in active users, successful tasks, and paid usage;

- Skill developers receive sustainable service income, CAA settlement comes from real choices rather than forced subsidies;
- Higher ratio CAA into vCAA;
- Reward compound interest is higher than withdrawal;
- Balance pool USDT reserve increased;
- New requirements cover reward transfer and lock-up release.

In this scenario, CAA quantity accumulation and price changes may form positive contributions at the same time, but it still does not constitute a guarantee.

Balance Scenario

- The new buy-in is close to the reward transfer;
- The market price fluctuates within the range;
- The main income of users is the change in quantity of CAA;
- Transfer fees and transaction costs may significantly affect actual results.

14.6 We are committed to making public economic indicators

- Actual tradable circulation volume;
- vCAA Pledge rate;
- Static/dynamic theoretical emissions;
- Actual weighted emissions;
- Withdrawal rate and compound interest rate;
- 30-day net buying and net selling;
- Balance pool USDT reserve rate;
- Top 10/50/100 address concentration;
- Estimated lifespan of reward pool;
- Lock the release amount in the next 30/90/365 days.

At the same time, we display the CARYA Agent product indicators and financial indicators separately:

- Daily/monthly active agents and active users;
- Next month retention rate of new users;
- Task submission volume, success rate, average completion delay and manual abort rate;
- Skill call volume, paid call volume, refund rate and developer net income;
- CAA, stablecoin and other assets' service settlement proportion;
- Model/API Cost, infrastructure cost and agreement service revenue;
- Manual confirmation rate of sensitive operations, number of security interceptions and disclosed events.

Chapter 15: Ecosystem Product Matrix and Future Infrastructure

15.1 Overall Positioning and CAA Integration

CARYA Labs uses CAA as the ecosystem value layer and organizes AI agents, a prediction market, a native public chain and a Meme launch platform as four ecosystem product directions. They create real utility entry points for CAA through product access, task settlement, developer incentives and service assurance, rather than reducing the laboratory narrative to one AI product.

Each application's specific functions, fees, Gas, settlement assets and incentive parameters will be disclosed separately when that product is formally released. Unreleased plans do not automatically create access, revenue, governance, price-support or other rights for current holders.

15.2 CARYA Agent: AI Agents and Developer Services

CARYA Agent gives individuals, teams and developers an executable work gateway that combines models, memory, skills, tools and external services. With explicit authorization, an agent can understand context, plan steps, invoke tools and deliver results.

CAA may support advanced-capability access, task services, developer-collaboration settlement, ecosystem incentives or service assurance. Verified usage, quality and safety records may inform incentives, and service providers may lock CAA under formal rules; the released rules define the final scope.

15.3 Prediction Market: Transparent Rules and Verifiable Outcomes

The planned prediction market creates markets around public events and uses trading prices to aggregate participants' views of possible outcomes. Each market should define its options, deadline, result source and dispute process in advance, then settle after a verifiable outcome is available.

CAA may support platform access, market-creation services, ecosystem incentives or service assurance. Participation methods, settlement assets, market scope, result sources, fees and compliance arrangements follow the official product documentation.

15.4 CARYA Chain: Future Native Public Chain

CAA is currently planned for deployment on BNB Smart Chain through compatible interfaces. CARYA Chain is future native execution infrastructure for CARYA Labs ecosystem applications, value settlement and verifiable collaboration, intended to provide one operating environment for AI, prediction markets, Meme launches and developer applications.

The future network may support smart contracts, data verification, tool interfaces and composable ecosystem services. CAA may take roles in Gas, application settlement, staking or network incentives, subject to a formal technical specification. Testnet, audit and production stages begin only after technical, security, migration and governance conditions are defined.

15.5 Meme Launch Platform: Issuance Services and Liquidity Collaboration

The planned Meme launch platform is an issuance tool for community assets and cultural IP, providing token-parameter and allocation-rule configuration, fair launch, participation access, time windows, liquidity-plan disclosure and ongoing ecosystem collaboration.

CAA may support platform services, ecosystem incentives or issuance assurance. Fees, issuance parameters, participation rules and launch timing follow the official release. Transparent rules and verifiable processes reduce information asymmetry but do not guarantee project quality or market outcomes.

15.6 Development Gates and Disclosure Boundaries

Future ecosystem development is advanced through verifiable gates rather than fixed dates or slogans: product specifications, permission boundaries, security testing, compliance assessment, independent audits, reserve impact and governance decisions. Before launch, each application should disclose CAA utility, fee routing, settlement assets, risk conditions and pause mechanisms.

Real demand from ecosystem applications may expand CAA utility, but does not guarantee token price, returns, buybacks or liquidity. CAA integration must be grounded in real use, transparent rules, asset segregation and verifiable data.

Chapter 16: Smart Contract Architecture

16.1 Module division

Module	Main responsibilities
CaaToken	Fixed upper limit assets, transfers, black hole records
StakingVault	CAA 1: 1 Exchange, vCAA Valid Principal, Rewards vCAA, Redemption and Solvency
RewardEngine	Static Reward vCAA, Dynamic Reward for Effective Cell Performance x Gear Daily Rate, Cycle Status
ReferralRegistry	Single parent referral tree, 256 direct address restriction, first binding 0.0005 BNB, acyclic validation
EpochSettlement	Merkle Root, Rewards vCAA, Redemption and Auto-Return Proof
FeeRouter	60 Day Redemption Fee, Balanced Pool/Operations 50% Distribution, CTC Balanced Pool/Community Incentive 2.5% Routing
CtcOrderBook	CTC/OTC Order Hosting, Matching, Order Withdrawal, Whitelisting Fee Free and Fee Settlement
BalanceVault	Balanced Pool CAA/U Assets, Four Door Quotes and Restricted Uses
BurnController	Black hole transfer, 0% — 100% synchronous destruction ratio, effective supply and double stop line
VestingVault	60-month linear release
MarketPolicyOracle	BAIQ four gates, δ working belt, quotation parameters and expiration date
AmmMigration	V2 Manual LP, LP Black Hole Attribution, Built-in Swap and CTC Shutdown Process
AgentRegistry	On-chain identification and status of agents, workspaces and developer identities
SkillRegistry	Skill manifest hash, version, publisher, permission summary and revocation status
AgentPaymentRouter	Multi-asset service metering, payment routing, fee disclosure and developer settlement
ServiceEscrow	Escrow, Confirmation, Refund and Dispute Status of Acceptable Tasks
AgentPolicyRegistry	Permission policy hash, restricted session authorization and revocation logging
GovernanceTimelock	Parameter changes, upgrade delays and execution

Agent reasoning, memory retrieval, model invocation and tool execution mainly run off-chain. Our on-chain module does not attempt to verify the "smartness" of the model, but only verifies identity, authorization, amounts, state transitions, time boundaries and verifiable receipts. We will not write original prompt words, complete conversations and sensitive data to the chain by default.

16.2 The upgradability boundaries we set

- We will make CaaToken maximum supply and genesis allocation non-upgradeable, or permanently turn off minting permissions;

- We are not allowed to increase the total amount of locked positions or change the established release curve;
- User principal ownership and confirmed rewards may not be arbitrarily overwritten by upgrades;
- CTC, settlement coordination and AI interface can be modularly upgraded, but must be audited, multi-signed and time-locked;
- CARYA Agent The runtime, model adapter and skill executor can be upgraded independently, but versions affecting wallet authorization, asset settlement or on-chain voucher format must be announced, tested and allow users to revoke old authorization;
- Before and after the upgrade, it must be ensured that the vault assets are consistent with the user's liabilities.

16.3 Permission roles

Role	Permissions	Constraints
DEFAULT_ADMIN	Management Roles	must be time lock/multi-signature, not ordinary single signature
PAUSER_ROLE	Pause affected modules	Do not withdraw treasury assets
SETTLEMENT_ROLE	Submit cycle settlement Root	Do not over-allocate the reward pool
MARKET_ROLE	Submit a BAIQ quote proposal	Restricted by Net U, Inventory Skew, Destruction Intensity, Inventory Depth, δ, Quota and Expiration Boundary
AGENT_REGISTRAR	Register or deactivate agent and skill versions	Do not grant asset permissions on behalf of the user
DISPUTE_ROLE	Handling managed service disputes	Subject to evidence, deadlines, multiple signatures and appeals process
TREASURY_ROLE	Execution of whitelist reserve purpose	Multiple signatures, quotas, time locks
UPGRADER_ROLE	Perform Approved Upgrade	Only time lock can be held
DAO_EXECUTOR	Execution governance passes proposal	Subject to governance processes and hard caps

We use the ERC-20, AccessControl, Pausable and ReentrancyGuard general modules provided by OpenZeppelin as the basis for implementation, and conduct independent testing and auditing of the combinatorial logic of this protocol.

16.4 Core Event

```
Stake(user, caaAmount, vcaaAmount, effectiveEpoch)
Unstake(user, grossAmount, netAmount, feeRate)
BindingFeePaid(user, referrer, bnbAmount)
RewardRootCommitted(epoch, merkleRoot, theoreticalTotal, alpha)
RewardClaimed(user, epoch, staticVcaa, dynamicVcaa)
RewardCompounded(user, amount, effectiveEpoch)
ReferralBound(user, referrer)
OrderPlaced(orderId, user, side, price, amount)
OrderFilled(orderId, counterOrderId, price, amount)
OrderCancelled(orderId, remainingAmount)
FeeRouted(asset, amount, balanceShare, communityShare, operationShare)
BurnSyncRatioChanged(oldRatio, newRatio, executionTime)
BurnTransferred(source, amount, effectiveSupply)
BurnStopTriggered(stopType, threshold)
BaiqQuoteCommitted(policyHash, delta, validFrom, validUntil)
AgentRegistered(agentId, owner, manifestHash, version)
SkillPublished(skillId, publisher, manifestHash, permissionHash, version)
SkillRevoked(skillId, version, reasonHash)
AgentAuthorizationGranted(user, agentId, scopeHash, expiry, limit)
AgentAuthorizationRevoked(user, agentId, scopeHash)
ServiceEscrowOpened(taskId, payer, provider, asset, amount, termsHash)
ServiceEscrowSettled(taskId, providerAmount, refundAmount, resultHash)
AmmMigrationScheduled(executionTime, reserveThreshold)
ParameterChanged(key, oldValue, newValue, executionTime)
```

16.5 Contract Invariants

We require auditing and formal testing to verify at least:

1. CAA Supply shall not exceed 310,000,000;
2. After turning off the minting permission, no role can be reopened;
3. The balance of pledge vault CAA shall not be less than the user's principal liability;
4. The actual reward in a single cycle shall not exceed the available balance of the reward pool;
5. The same settlement certificate cannot be received repeatedly;
6. There is no self-binding or looping in the recommendation relationship;
7. The direct subordinates of each address shall not exceed 256;
8. The transfer rate can only be reduced gradually over time and shall not be less than 10%;
9. Each transfer fee is strictly 50%/50% routing;
10. CTC/OTC orders cannot be overtaken or custodian assets released repeatedly; balanced pool whitelisting orders must be free of pending order fees and closing fees;
11. The synchronous destruction ratio can only be adjusted within the range of 0% — 100%, and the corresponding destruction action cannot continue to be deducted after any stop line is reached;
12. AI quotation cannot cross the hard boundary on the chain;
13. Suspended characters cannot bypass treasury spending rules;
14. The cumulative release of locked positions shall not exceed 10,000,000 CAA;

15. Agents and skills must not continue to submit executable asset operations after the authorization expires, is revoked, or exceeds the limit;
16. The total amount of service custody settlement shall not exceed the amount locked by the user;
17. The same task receipt, authorization nonce or settlement voucher cannot be reused;
18. On-chain authority credentials must not contain wallet private keys, original conversations, or plain text sensitive data.

Chapter 17: Security Architecture and Audit

17.1 Smart Contract Security

We will focus on checking:

- Reentrancy attack;
- Precision and rounding;
- Signature replay;
- Merkle Proof Duplicate collection;
- Recommendation relationship cycle;
- Partial transaction of the order was incorrect;
- Malicious ERC-20 return value;
- Privilege escalation and proxy storage conflict;
- Oracle/AI parameter manipulation;
- MEV, front-running and clip trades;
- Pause and resume logic error.

We follow the check-status update-external interaction sequence in our implementation, and use reentrancy protection, function-level suspension, asset whitelisting, signature domain isolation, and strict balance invariants. We will also conduct code review, testing, auditing and design a fail-safe mode for the production contract based on the Solidity official document.

17.2 Agents and Data Security

We connect model output to tools and external services via CARYA Agent, so we will protect model calls, data flow, and execution links at the same time. We focus on checking:

- Prompt injection induces the agent to ignore user intent or leak context;
- Malicious web pages, files, emails, skills or plug-ins implement data external transmission;
- Supply chain integrity of third-party models, API, dependency packages and update channels;
- Key, token, session, memory or log was read without authorization;
- Model hallucination leads to wrong call, wrong recipient or wrong on-chain parameters;
- The permission inheritance between multiple agents is unclear, resulting in permission expansion;
- Automated tasks continue to execute expired instructions after environment changes;
- Developers forge calls, results or quality indicators to defraud settlement and incentives.

We adopt defense in depth: model and executor isolation, skill list signature, dependency locking, sandbox, network egress control, secret information vault, minimal context, structured tool parameters, address and contract whitelist, single/cycle quota, manual approval, rate limit, anomaly detection, revocable authorization

and immutable audit log.

We do not allow agents to have direct access to wallet mnemonics or private keys. Before sensitive tools go online, we will complete independent threat modeling, red team testing and minimum privilege review; when the security policy fails, we will stop the corresponding tools, but will not freeze users' normal access to assets on our own chain.

17.3 Security Control Tools

- Reserve rate warning;
- Large order limit;
- Single-day balance pool fund usage limit;
- Quotation validity period;
- Abnormal fluctuation fuse;
- Modular suspension;
- Multiple signatures and time locks;
- Independent on-chain monitoring;
- Bug Bounty;
- Incident response plan;
- AMM Pre-migration stress testing.
- Agent authority center and one-click revocation;
- Skill signature, version lock and malicious plug-in removal;
- Manual approval and expenditure limit for sensitive tools;
- Prompt injection, data outgoing and abnormal call monitoring;
- Degrade and isolation switches for models, tools, and external API;
- Security incident announcement, user notification and authorization rotation process.

17.4 Audit Scope

We will entrust an independent third party to conduct an audit, and the audit scope will at least cover:

- CAA Token;
- StakingVault/vCAA;
- RewardEngine and settlement certificate;
- ReferralRegistry;
- FeeRouter;
- CTC OrderBook;
- BalanceVault;
- BurnController;
- VestingVault;

- MarketPolicyOracle;
- AgentRegistry, SkillRegistry and AgentPolicyRegistry;
- AgentPaymentRouter and ServiceEscrow;
- CARYA Agent Identity, session, memory, permissions, sandbox and key isolation;
- Models, Skills, Plugins, Updates and Third Party API Supply Chain;
- AMM Migration module;
- Manage, upgrade and suspend permissions.

Chapter 18: Transparency and Governance

18.1 On-chain transparency

We will continue to disclose in the protocol data center:

- Maximum supply, total supply on the chain and effective supply;
- Black hole balance;
- Tradable circulation;
- vCAA Pledge amount;
- Static and dynamic rewards to be claimed vCAA;
- Reward pool balance and weighting coefficient;
- Balance pool assets and historical changes;
- CTC/OTC orders, deals, valid fee-inclusive prices and balance pool whitelisting status;
- Destruction source, burnSyncRatio and stop line status;
- Technical and operational share release;
- Multi-sign members, thresholds and time locks;
- AI model version and quotation strategy hash;
- CARYA Agent Runtime, model provider, skill and plug-in version;
- Agent service price, supported settlement assets and fee allocation;
- Aggregation indicators such as active users, successful tasks, paid calls, refunds and developer income;
- Sensitive permission policies, security incidents, affected versions and repair status;
- Parameter change history.

We will disclose product indicators in an aggregated and desensitized manner, and will not disclose user conversations, personal files, private tasks, wallet identity associations or other sensitive data. We will not expand the original scope of authorization through any data panel on the grounds of "transparency".

18.2 Our progressive governance path

Initial stage

- Core team multi-sign;
- Key parameter time lock;
- Emergency suspension of role separation;
- All operation chains leave traces.

Transitional Phase

- Community proposal discussion;

- Public simulation of parameter changes;
- DAO Signal voting;
- Multiple signatures are executed based on the voting results.

Mature stage

- On-chain proposals and voting;
- Timelock Automatic execution;
- Reserve expenditure budget;
- AI model and AMM parameters are included in governance;
- Agent service rules, skill market policies and developer budgets are included in the management;
- Emergency powers are subject to double veto and expiration limits.

18.3 Governable parameters and hard boundaries

We plan to permanently fix:

- Maximum supply;
- Total technical and operational lockup;
- User assets and reward records that have occurred;
- The black hole balance is not recoverable;
- The vCAA redemption fee is based on the principle of 50% of the balanced pool and the operating address; the fee of 5% on each side of CTC is fixed and routed according to the balanced pool 2.5%/community incentive 2.5%;

We allow governance within established hard caps:

- BAIQ four-door control, daily reference price overhaul δ and balance pool single day quota;
- Single-day quota of the balancing pool;
- AMM Start threshold;
- AMM Handling fee;
- Meltdown threshold;
- Settler and challenge parameters;
- Reserve asset whitelist;
- Settlement assets, protocol service fee limit and developer incentive budget supported by the intelligent agent;
- Skill release standards, margin range, dispute period and sensitive authority level.

We will not include users' original data, wallet private keys and specific task contents into DAO governable assets. Any governance vote shall not retroactively read user private data, nor shall it revoke the user's right to revoke agent authorization.

Chapter 19: Our Implementation Roadmap

We will promote the construction, testing, online and governance transfer of the Walnut Network and CARYA Agent in accordance with the following stages. Each stage is subject to actual completion and public announcements.

Phase 0: Mechanism Freezing and Simulation

- Complete the white paper and parameter sheet;
- Establish a reward pool life model;
- Simulate user growth, compound interest, withdrawals and price pressure;
- Freeze destruction sources, 0% — 100% synchronous destruction ratio and double stop line realization;
- Determine the multi-signature and governance structure;
- Complete CARYA Agent product definition, permissions model, runtime prototype, and threat modeling.

Phase 1: Testnet and Audit

- Deploy CAA, vCAA, rewards, recommendations, CTC, balancing pool and staking modules;
- Perform unit testing, fuzz testing, property testing and stress testing;
- Complete at least one independent third-party audit;
- Establish a public testnet data panel;
- Release CARYA Agent closed beta, verifying sessions, memory, model routing, skills sandbox and manual approval.

Phase 2: CTC/OTC mainnet launch

- Creation allocation and lock-up;
- Initial 0 market circulation;
- Open CTC/OTC Buy Order, Balanced Pool Whitelisting and BAIQ Gated Progressive Release;
- CAA 1: 1 redemption of vCAA, vCAA rewards every 24 hours, automatic replay, redemption and referral system;
- Continuous disclosure of market and reserve data;
- Open personal agent beta, the first batch of official skills and user permissions center.

Phase 3: Market Maturity Assessment

- Monitoring balance pool USDT;
- Assess circulation, holders and concentration;

- Conduct third-party review of AI quotation;
- Announcement of AMM migration standards and stress testing;
- Release developer SDK, skills list standards, test environment and signature release process;
- Trial operation of skills market and multi-asset service payment, independent disclosure of product income and agreement rewards.

Phase 4: V2 LP with built-in Swap

- Governance Approval LP parameter;
- CTC Stop adding new orders;
- Manually create CAA/U LP as per V2 process;
- Open built-in Swap; BTC, ETH, BNB and other mainstream currencies obtained therein are not transferable;
- Disclose LP and reserve data, and transfer LP ownership and management rights to the black hole address;
- Open AgentPaymentRouter, task hosting, and CAA service settlement within the audit and governance boundaries.

Phase 5: Reserves and DAO

- Expand ecological reserves;
- Establish multi-chain asset certificate;
- Gradually transfer governance authority;
- Open ecological applications and developer interfaces;
- Developer program to expand multi-agent collaboration, third-party professional agents and community governance.

Phase 6: Ecosystem Applications and Native-Chain Evaluation

- Expand CARYA Agent advanced capabilities, task services and developer-collaboration interfaces;
- Release product specifications, result-source or issuance rules, dispute processes and controlled tests for the prediction market and Meme launch platform;
- Publish the CARYA Chain technical specification, testnet plan, security audit, asset-migration and governance plan before mainnet launch;
- Disclose CAA utility, fee routing, reserve impact and pause boundaries for each application.

Appendix A: Current parameter table

Parameters	Current value
Maximum Supply	310,000,000 CAA
Ecological reward pool	290,000,000 CAA
Balance Pool	10,000,000 CAA
Technical and operational lockup	10,000,000 CAA
Lock-up period	60 months
Minimum pledge	100 CAA
Pledge/Exchange Incremental Units	1 CAA
Reward cycle	24 hours
Maximum number of direct addresses	256
Initial transfer fee	30%
Decreasing transfer fee	5 percentage points every 60 days
Minimum transfer fee	10%
Transfer fee allocation	Balance pool 50% / Operation address 50%
CTC/OTC pending order fee	0.0001 BNB
CTC Buyer's Closing Fee	5%
CTC Seller closing fee	5%
CTC Cost allocation per side	Balance Pool 2.5% / Community incentive 2.5%
Effective supply stop line	100,000,000 CAA
Reward pool destruction stop line	10,000,000 CAA
AMM Start USDT Threshold	TBD
AMM Handling fee	TBD
AI maximum quotation deviation in a single period	TBD
CARYA Agent Launch model and channel	TBD
Agent service settlement assets	CAA / USDT / Other Governance Approved Assets (Scope TBD)
Smart Agent Agreement Service Fee Upper Limit	TBD
Default policy for sensitive operations	Manual confirmation
Skill Release Signature and Security Rating	TBD
Single and periodic asset authorization quota	User settings/protocol upper limit TBD
Multi-signature threshold	TBD
Manage time lock	TBD
Binding Fee	0.0005 BNB
CTC/OTC Minimum Pending Orders	100 CAA; integer multiplied by 1 CAA
Balance Pool Whitelisting	Waived Hanging Order Fee and Closing Fee
BAIQ Opening Reference Price	1.50 U/1 CAA
BAIQ daily reference price up revision δ	Regular Day 0.10% — 0.20%; Strong Absorption 0.30% — 0.40%; From Extreme Day 0.50%; Ungated 0
Reward Pool Sync Destruction Ratio	Default 100%; adjustable in 0% — 100%
V2 LP Ownership and Administrative Permissions	Go to black hole address
BTC/ETH/BNB obtained with built-in Swap	Non-transferable